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Cellular Automata for Two Lane Traffic Flow

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Abstract

Key Words:

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1 Introduction

1.1 Real-world Context

Urban Traffic Congestion is a pervasive issues across a range of different demographics, whether it is in a developing or developed country. Traffic Congestion has always been a significant concern in traffic control mechanisms caused by an unbalanced demand-supply situation that directly affects the problem of large fuel consumption and greenhouse emission. Congestion leads can lead to a range of issues such as traffic jams, unhealthy air quality, vibrations, noises and accidents which are significant problems for the environment and human health [1].

In the UK, this issue is particularly prevalent. In the 2019, UK drivers lost an average of 115 hours each to road traffic congestion [2].

To reduce congestion, we should look at optimising traffic flow, this can be done through the use of traffic lights, by employing various strategies we can maximise the flow of traffic and reduce congestion. Since traffic lights are already widely used, it is important to understand how they can be optimised to improve traffic flow and reduce congestion allowing for minimal costs.

1.2 Why discrete cellular automata

We can leverage the use of cellular automata to model traffic flow, this is because they are a powerful tool for simulating complex systems and can capture the dynamics of traffic flow in a simple and efficient way. Cellular Automata also allow us to assess the impact of different traffic light strategies on traffic flow, and can help us to identify the most effective strategies for reducing congestion.

Traffic cellular automata models have been widely used in traffic flow research, and have been shown to be effective in capturing the dynamics of traffic flow and congestion. Examples of cellular automata models include the Nagel-Schreckenberg (NaSch) model and the Biham-Middleton-Levine (BML) model [3, 4].

For this study, we will use the NaSch model due to its simplicity and more clearly explaining the effects on traffic lights in the system. With this model we can easily consider the effects of multiple traffic lights on the system as well as the effects of different traffic light strategies on traffic flow and congestion. Additionally we can also consider how a second lane can affect the traffic flow and congestion in the system with different vehicle types taken into consideration.

1.3 Aims and Research Questions

In this report, we will explore two different contexts of traffic flow, exploring how different traffic light layouts effect the system. The first context will be a single lane system, where we vary traffic light number and traffic light strategies [3]. The second one will be a two lane system, where will explore the effect of varying traffic light number and traffic light strategies with the introduction of different vehicle types on the system [5, 6]. The unifying question across these two is light configuration and strategy, we want to understand how different traffic light configurations and strategies can affect traffic flow and congestion in both single lane and two lane systems. The two geometries are shown in Figure 1.

We will address four research questions:

1. How does single-lane through-flow scale with the number of traffic lights, the cycle length, the green fraction and the inter-light phase offset?
2. When does green-wave coordinate outperform synchronised lights and why?
3. Does symmetric lane changing complement the green-wave or fight it?

2 Background

2.1 Continuum vs Discrete Traffic Flow

Analytical traffic models can be either continuum or discrete. Early examples of traffic flow models were based on continuum mechanics. Continuum models treat traffic as a continuous fluid, where instead of modelling individual vehicles, the vehicle density, mean velocity and flow are all modelled as continuous functions of space and time. Lighthill and Whitham represented the road as a one-dimensional fluid with flow q (vehicles per hour), concentration k (vehicles per mile) and posited as the fundamental hypothesis of their theory that the flow q is a function of k alone [7]. The continuum models are useful for understanding traffic flow at a macroscopic scale, formation of traffic jams and the effects of traffic lights but cannot capture the microscopic behaviour of individual vehicles and the interactions between them. Discrete models on the other hand address this weakness by treating everything as a finite integer such as space, time, velocity and the vehicles themselves, as we discuss next.

2.2 The NaSch cellular automaton

Nagel and Schreckenberg proposed a stochastic discrete cellular automaton where a road is represented as a one-dimensional lattice of cells, each of which can be occupied by a single vehicle or be empty. Each vehicle carries an integer velocity v in the range $\{0, 1, \dots, v_{\max}\}$ and the system is updated in discrete time steps according to a set of rules that govern the movement of the vehicles. The rules are as follows:

1. **Acceleration:** If the velocity of a vehicle is less than the maximum velocity $v_{\max}$, and if the distance to the next car is larger than $v + 1$, the velocity is increased by 1, $[v \rightarrow v + 1]$.
2. **Deceleration:** If a vehicle at cell i sees the next vehicle at cell $i + j$ (with $j \leq v$), it reduces its speed to $j - 1$, i.e. $[v \rightarrow j - 1]$.
3. **Randomisation:** With probability p , the velocity of each vehicle (if greater than zero) is decreased by one $[v \rightarrow v - 1]$.
4. **Movement:** Each vehicle is advanced v cells.

Here p is the probability of random deceleration, which is a key parameter that controls the behaviour of the system. Through the four rules above, the general properties of single lane traffic are modelled on the basis of integer valued probabilistic cellular automaton rules. This simple model is effective in capturing the nontrivial and realistic behaviour of traffic flow. Step 3 is particularly important for simulating realistic traffic flow, without it the dynamics is purely deterministic, and from any initial configuration the system relaxes to a stationary pattern shifted backwards one cell per timestep [3, 8]. The model also accounts for natural velocity fluctuations due to human behaviour or due to varying external conditions. The four steps above are unchanged across the single lane and two lane scenarios.

2.3 Two-Lane Extensions

Rickert et al. proposed a two-lane extension to the NaSch model, where vehicles can change lanes according to a set of rules that depend on the local traffic conditions. The lane-change sub-step runs before the NaSch update, with both sub-steps applied as a strict parallel update against the configuration at the start of each time step [5].

Chowdhury et al. extended Rickert's work to include two different types of vehicles, slow (e.g. a truck) and fast (e.g. a car), and proposed two variants of the lane change rule: a *symmetric* variant in which the same rules apply for both lanes, and an *asymmetric* variant modelled after the German Autobahn convention where the right (slow) lane is preferred and the left (fast) lane is used for overtaking only [6]. In this report we will explore both variants of the lane change rule.

For the symmetric version, the rules for the parallel updating of the speeds and positions of the vehicles are as follows: first, each vehicle is considered for the possibility of a lane changing and if the rules allow the lane change, the changes are implemented; second the speed and position of each vehicle is updated following the rules for the forward movement in the current lane. A vehicle changes lane with probability p_{lc} provided:

1. the gap in front in the current lane is less than $v + 1$

2. the gap in front in the other lane is larger than the gap available in front in the current lane
3. the nearest-neighbour site in the other lane is empty
4. the maximum possible speed of the vehicle behind in the other lane is smaller than the gap in between

In the asymmetric version, the change from the slow lane to the fast lane uses the four conditions of the symmetric version, but the change from the fast lane back to the slow lane is weaker: a vehicle returns to the slow lane as long as conditions (3) and (4) above are satisfied and the slow-lane gap is sufficient, without requiring (1) and (2) to be satisfied.

The rules for updating the speeds and positions of the vehicles in the current lane are identical to those in the NaSch single lane model in Section 2.2 [6]. Lane-changing alters the design space for light optimisation, since cars can route around a queue at one light by switching lane.

2.4 Optimising Signalised Systems

Traffic lights are a common feature of urban traffic networks, and their timing can have a significant impact on traffic flow and congestion. Brockfeld et al. combined the basic ideas of the Biham-Middleton-Levine (BML) model for city traffic with the NaSch model for highway traffic to create a square lattice geometry [9]. In this model, each cell can be occupied by a vehicle moving in one of two directions, and the traffic lights at the intersections control the flow of vehicles through the lattice. In this paper various strategies for optimising the traffic light timings are explored:

1. **Synchronised traffic lights** - all lights switch simultaneously, producing strong throughput oscillations as the cycle length is varied.
2. **Green wave strategy** - the phase offset between successive lights is set so that a vehicle cruising at the free flow speed $v_{\text{free}} = v_{\text{max}}p$ arrives at each light just as it turns green, allowing it to pass through multiple lights without stopping.
3. **Random offset strategy** - each light's offset parameter is chosen from a uniform distribution over the interval $\{0, \dots, 2T\}$

Green wave gives much improvement to the flow of traffic in comparison to the synchronised strategy. Random offset outperforms the synchronised strategy at low densities for small cycle times, and at high densities for all cycle times [9].

3 Model Description and Boundary Form

3.1 Model Assumptions

3.2 Model Scaling and Interpretation

The model can be scaled so that each car occupies around 7.5 m of road which is the length of one cell. As the average velocity of cars in free traffic is 4.5 cells per time step it should correspond to around 120 km/h (in Germany), which is a reasonable speed for free traffic on a motorway, naturally we arrive at a time for iteration of

$$7.5 \frac{\text{m}}{\text{cell}} \times 4.5 \frac{\text{cells}}{\text{time-step}} / (120/3.6) \frac{\text{s}}{\text{m}} \approx 1.0 \text{ s} \quad (1)$$

[3]. As this model is scalable to different scenarios there is not a need to non-dimensionalise the model, we can simply use the above scaling to interpret the results in real world terms

3.3 Road Geometries and Light Configuration

The two road geometries studied in this report are shown in Figure 1.

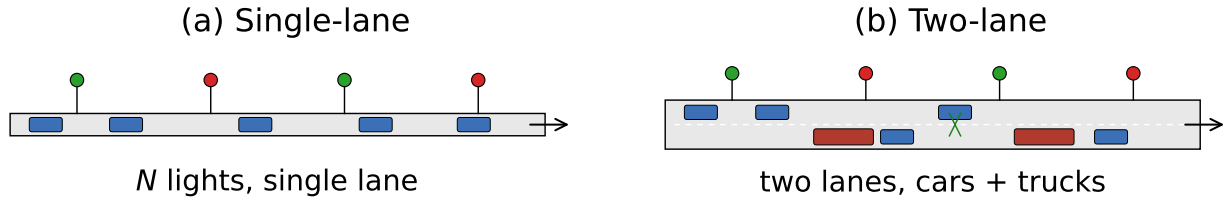


Figure 1: The two road geometries studied in this report. (a) Single-lane road with N traffic lights, cars travel left-to-right. (b) Two-lane road with mixed vehicle types (cars in blue, trucks in red) and N synchronised lights, the green arrow marks a candidate symmetric lane change. Cell counts, light spacing, and cycle parameters are set in the corresponding model section. Schematics are illustrative

3.4 Single Lane Lattice and State

3.5 NaSch Update Rules

3.6 Boundary Conditions

3.7 Traffic Lights as Obstacles

3.8 Two-Lane Extension and The CWS Lane-Change Rule

4 Numerical Method and Validation

4.1 Parallel-Update Implementation

4.2 Measurement Protocol

4.3 Validation Gates

4.4 Reproducibility

5 Baseline Results

5.1 No-Lights Baseline

5.2 Single-Lane Lights

5.3 Two-Lane Lights

5.4 Cross-Context Synthesis

6 Sensitivity Analysis

6.1 Methodology

6.2 Tornado Plot Ranking

6.3 Top-3 Response Curves

6.4 Joint Sweep

6.5 Convergence Diagnostic

6.6 Worst-Case Robustness Bound

6.7 Robustness Statement

7 Conclusion

References

- [1] U. Jilani, M. Asif, M. Y. I. Zia, M. Rashid, S. Shams, and P. Otero, “A Systematic Review on Urban Road Traffic Congestion,” *Wireless Personal Communications*, vol. 140, pp. 81–109, Jan. 2025.
- [2] B. Pishue, “2020 INRIX Global Traffic Scorecard,”
- [3] K. Nagel and M. Schreckenberg, “A cellular automaton model for freeway traffic,” *Journal de Physique I*, vol. 2, pp. 2221–2229, Dec. 1992.
- [4] O. Biham, A. A. Middleton, and D. Levine, “Self-organization and a dynamical transition in traffic-flow models,” *Physical Review A*, vol. 46, pp. R6124–R6127, Nov. 1992.
- [5] M. Rickert, K. Nagel, M. Schreckenberg, and A. Latour, “Two lane traffic simulations using cellular automata,” *Physica A: Statistical Mechanics and its Applications*, vol. 231, pp. 534–550, Oct. 1996.
- [6] D. Chowdhury, D. E. Wolf, and M. Schreckenberg, “Particle hopping models for two-lane traffic with two kinds of vehicles: Effects of lane-changing rules,” *Physica A: Statistical Mechanics and its Applications*, vol. 235, pp. 417–439, Feb. 1997.
- [7] M. J. Lighthill and G. B. Whitham, “On kinematic waves II. A theory of traffic flow on long crowded roads,” *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, vol. 229, pp. 317–345, May 1955.
- [8] M. Schreckenberg, A. Schadschneider, K. Nagel, and N. Ito, “Discrete stochastic models for traffic flow,” *Physical Review E*, vol. 51, pp. 2939–2949, Apr. 1995.
- [9] E. Brockfeld, R. Barlovic, A. Schadschneider, and M. Schreckenberg, “Optimizing traffic lights in a cellular automaton model for city traffic,” *Physical Review E*, vol. 64, p. 056132, Oct. 2001.